

Muhammad Uzair Khattak

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[LinkedIn portfolio](#)

EDUCATION

Doctor of Philosophy - PhD, Computer Vision

Swiss Federal Technology Institute of Lausanne (EPFL)

Sept. 2024 – current

- PhD student (EDIC Fellowship) supervised by [Prof. Amir Zamir](#) and [PD. Dr. Federico Tombari](#)
- Current research focus: Any-to-any video modeling, world models in multimodal spaces, multi-task learning

Master of Science, Computer Vision

Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi

August. 2021 – June 2023

CGPA 4.00/4.00

- Graduate research student supervised by [Dr. Salman Khan](#) and [Dr. Fahad Khan](#)
- Research thesis: "Transferability of vision-language models with prompt learning for image and video tasks"
- Major courses: Human and Computer Vision, Machine Learning, Visual Object Recognition and Detection

Bachelor of Electrical Engineering

School of Electrical Engineering and Computer Science (SECS)

National University of Sciences & Technology (NUST), Islamabad, Pakistan

Sept. 2017 – June 2021

CGPA 3.84/4.00

- Undergraduate student supervised by [Dr. Hassan Aqeel Khan](#) and [Dr. Faisal Shafait](#)
- Final year thesis: "Low-Cost Whole Slide Image Scanner with Deep Learning Applications" (Awarded Rector's Gold medal)
- Major courses: Signal Processing, Embedded Systems, Microprocessor Systems, Probability & Statistics, Data Structures

RESEARCH INTERESTS

- Multimodal video modeling, and its implications for world modeling and long-horizon prediction tasks
- Pretraining vision-language models, reasoning in video LMMs, and test-time optimization
- Transfer learning techniques with prompt learning for the robustness and generalization of vision-language models

PUBLICATIONS

* indicates joint first authors, + indicates my role as co-mentor

Any-to-Any Video Modeling in Multimodal Abstract Spaces

Muhammad Uzair Khattak, Won Jun Kim, Reza Abbassi, Michael Murphy, Amir Zadeh, Chuan Li, Oguzhan Fatih Kar, Muhammad Ferjad Naeem, Andrei Atanov, Roman Bachmann, Federico Tombari, Amir Zamir

May 2026 (Under Review)

Complex Video Reasoning and Robustness Evaluation Suite for Video-LMMs ([Paper](#)) ([Code](#))

Muhammad Uzair Khattak, Muhammad Ferjad Naeem, Jameel Hassan, Federico Tombari, Fahad Khan, Salman Khan

May 2024 (CVPRW-2025)

Learning to Prompt with Text Only Supervision for Vision-Language Models ([Paper](#)) ([Code](#))

Muhammad Uzair Khattak, Muhammad Ferjad Naeem, Muzammal Naseer, Luc Van Gool, Federico Tombari

Dec 2024 (AAAI-2025)

Align Your Prompts: Test-Time Prompting with Distribution Alignment for Zero-Shot Generalization

Jameel Hassan*, Hanan Ghani*, Noor Hussein*, **Muhammad Uzair Khattak**+, Salman Khan, Fahad Khan

April 2023 (NIPS-2023)

Video-FocalNets: Spatio-Temporal Focal Modulation for Video Action Recognition ([Paper](#)) ([Code](#))

Syed Talal Wasim*, **Muhammad Uzair Khattak***, Muzammal Naseer, Salman Khan, Mubarak Shah, Fahad Khan

March 2023 (ICCV-2023)

Self-regulating Prompts: Foundational Model Adaptation without Forgetting ([Paper](#)) ([Code](#))

Muhammad Uzair Khattak*, Syed Talal Wasim*, Muzammal Naseer, Salman Khan, Ming-Hsuan Yang, Fahad Khan

March 2023 (ICCV-2023)

Fine-tuned CLIP models are efficient video learners ([Paper](#)) ([Code](#))

Hanoona Rasheed*, **Muhammad Uzair Khattak***, Muhammad Maaz, Salman Khan, Fahad Khan

Feb 2023 (CVPR-2023)

MaPLe: Multi-modal Prompt Learning ([Paper](#)) ([Code](#))

Muhammad Uzair Khattak, Hanoona Rasheed, Muhammad Maaz, Salman Khan, Fahad Khan

Feb 2023 (CVPR-2023)

Bridging the Gap between Object and Image-level Representations

for Open-Vocabulary Detection ([Paper](#)) ([Code](#))

Hanoona Rasheed*, Muhammad Maaz*, **Muhammad Uzair Khattak**, Salman Khan, Fahad Khan

May 2022 (NIPS-2022)

Investigating and Improving Common Loop Closure Failures in Visual SLAM ([Paper](#))

Saran Khaliq, Muhammad Latif Anjum, Wajahat Hussain, **Muhammad Uzair Khattak**, Momen Rasool

July 2022 (Autonomous Robots)

EXPERIENCE

Research Associate II

Mar 2024 – current

Intelligent Visual Analytics Lab (IVAL), MBZUAI, Abu Dhabi

Research topics: Video Large Multimodal Models (LMMs), Prompting techniques for LMMs, Contrastive Vision-Language Models

- Investigating and bench-marking hallucination in Video LMMs
- Training free-prompting techniques for steering LMM's behavior (self-reasoning)
- Text-only supervision for Vision-Language models with prompt learning
- Co-supervising new master students at IVAL lab

Graduate Research Assistant

Aug 2021 – July 2023

Intelligent Visual Analytics Lab (IVAL), MBZUAI, Abu Dhabi

Main research topics: Object detection with transformers, Multi-modal modals, Prompt learning

- Explored CLIP adaptation for action recognition, object recognition and open-vocabulary object detection
- Worked on transformers based detectors including DETR and MDETR for generic and class-agnostic object detection
- Co-mentored new master students at IVAL lab, research outcomes resulted in a NeurIPS'23 paper
- Teaching assistant for Machine Learning course (ML701) and lab instructor for UGRIP internship program at MBZUAI

Undergraduate Research Assistant

May 2019 – May 2021

Signal Processing and Machine Learning (SIGMA) Lab in collaboration with TUKL

R&D Center, SEECs, NUST, Islamabad, Pakistan

Main research topics: Medical Imaging, Signal processing, AI on Edge devices, Embedded systems

- Optimization of AI models (YOLOv4, ResNets) with TensorRT on edge devices including Jetson Nano and Jetson TX2
- Real-time auto-focusing with Laplacian filter as focus metric. Integrated algorithm with stepper motors circuitry
- Segmentation of keratin pearl and epithelium tissues using UNET in Whole Slide images
- Participant in international AI competition "Plant Pathology 2020 - FGVC7, CVPR-2020" ([GitHub](#))
- Teaching assistant for Complex Variables and Transforms course (MATH-232)

Artificial Intelligence Intern

Oct. 2020 – April 2021

i engineering Group, Islamabad, Pakistan

- Worked on project "Base station security (BTS) system", created dataset at BTS sites in Pakistan for intruder detection
- Implemented quantized MobileNet model in PyTorch on NVIDIA JETSON Nano 2GB developer kit
- Developed RESTFUL API using Flask and deployed on Heroku web server
- Integrated AI system with i engineering servers and IP camera to provide real time intrusion detection
- Automated the pipeline (updating and shutting down AI algorithm remotely, fetching debug logs) using shell scripts

PROJECTS

Transformers Transforming Vision

Sept 2021 – Dec 2021

- Explored state of the art vision transformers for image recognition including ViTs, DeiTs and T2T-ViTs
- Performance scalability comparison of ViTs and DeiTs with CNNs (ResNets) w.r.t pretraining dataset size
- Benchmarking of ViTs and ResNets on downstream datasets CIFAR10, CIFAR100 and CUB-200 ([GitHub](#))

Understanding the limitations of CNN based pose regressions models for Scene Localization

Oct. 2020 – March 2021

- Explored DL based approaches in Scene localization problem for autonomous robots
- Used custom dataset "1-KM" to understand the drift of CNN pose regressors MapNet and PoseNet in large scenes
- Implementation of visualization software for converting 3-D scene poses to 2-D image projections on RGB images using camera poses and 3D point cloud data

TECHNICAL SKILLS

Languages: Python, C/C++, MATLAB, Assembly and embedded C, SQL, JavaScript, HTML/CSS

Programming: PyTorch, fast.ai, Keras, Latex, TensorRT, Brevitas, Flask, Tkinter, Scikit-learn, Hugging-Face

Tools: Linux, PyCharm, VS code, Git, QuPath, Google Cloud Platform, Heroku, tableau, AUTOCAD

HONOURS AND AWARDS

- MBZUAI graduate studies scholarship holder
- Second [position](#) in RTA Transport Hackathon 2022, organized by Dubai Roads and Transport Authority
- First [position](#) in ADAFSA Agrithon 2021, organized by Abu Dhabi Agriculture and Food Security Authority
- Third position in Hack for Space Hackathon 2021, organized by G42 and coders(hq), UAE
- Winner of the MBZUAI Dogs vs. Cats challenge, 2021 ([report and github repository](#))
- Among the first batch of NVIDIA certified AI Jetson specialists in the world
- Featured in official blog of NVIDIA regarding acknowledgment of AI skills on NVIDIA embedded systems ([Blog](#))
- 3rd time recipient of NUST high achievers award for achieving 4.00/4.00 GPA in three semesters

ACADEMIC SERVICES

- Attended CVPR-2023 conference in Vancouver, Canada.
- Serving as a reviewer at NeurIPS-2023, ECCV-2024, ICCV-2023, ICML-2024, CVPR-2023/2024 & TPAMI
- Served as a volunteer for [Vision Transformers: Theory and applications](#) workshop at ACCV-2022 and NeurIPS-2022
- Attended NeurIPS-2022, CVPR-2022 & NVIDIA GTC 2021 conferences virtually
- Workshop presenter on "Deep Learning using PyTorch", talk organized by IEEE SEECS, NUST ([Recorded session](#))
- Technical trainer at "AI at the Edge" workshop, organized by AI Lounge, Pakistan
- Mentor and project evaluator for new batch of SIGMA lab interns, NUST, Pakistan

EXTRACURRICULAR ACTIVITIES

- Team captain of SEECS, NUST table tennis team
- Mentor at ACM Peer-to-Peer Mentorship program
- Organized "Mall of Humanity" project to distribute clothes to deserving families in the backward areas of Pakistan
- Co-led community service program "Meet the Seniors", aiming to connect high school students with undergraduate students of different fields and universities for face to face mentor-ship sessions

OTHER INTERESTS

- Multiplayer Computer Games, Traveling, Watching movies, Table Tennis, Football, Cricket